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Directionality of Halogen-Bonds: Insights from 2D Energy Decomposition Analysis

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Halogen bonds are typically observed to have a linear arrangement with a 180° angle between the nucleophile and the halogen bond acceptor X–R. This linearity is commonly explained using the σ-hole model, although there have been alternative explanations involving exchange repulsion forces. We employ two-dimensional Distortion/Interaction and Energy Decomposition Analysis to examine the archetypal $H_3N\cdots X_2$ halogen bond systems. Our results indicate that although halogen bonds are predominantly electrostatic, their directionality is largely due to decreased Pauli repulsion in linear configurations as opposed to angled ones in the I_2 and Br_2 systems. As we move to the smaller halogens, Cl_2 and F_2, the influence of Pauli repulsion diminishes, and the energy surface is shaped by orbital interactions and electrostatic forces. These results support the role of exchange repulsion forces in influencing the directionality of strong halogen bonds. Additionally, we demonstrate that the 2D Energy Decomposition Analysis is a useful tool for enhancing our understanding of the nature of potential energy surfaces in noncovalent interactions.

Introduction

Halogen bonds, characterized by the attractive forces between halogen atoms and nucleophiles, have found widespread applications in diverse scientific disciplines. The concept of halogen bonding emerged as an extension of hydrogen bonding and has its roots in the early 20th century.[1] However, it was in the 21st century that researchers began to fully appreciate the unique role of halogen bonds in molecular recognition,[2] crystal engineering,[3] supramolecular chemistry,[4] organocatalysis,[5] and drug design.[6]

Despite their increasing application in applied sciences, foundational research into the basic understanding of halogen bonds continues. Two primary aspects of halogen bonds are under investigation: the nature of the interaction[7] and their directionality.[8] Halogen bonds typically exhibit a linear geometry, characterized by a 180° angle between the nucleophile and the halogen bond acceptor X–R. This linear configuration is often explained using the σ-hole model,[7b,8d, 9] which proposes a region of partially positive charge on the halogen capable of interacting with a nucleophile. In X–R halogen bond acceptors this σ-hole aligns with the X–R bond axis, resulting in a linear bond configuration. The σ-hole model suggests that the directionality of these bonds is fundamentally an electrostatic phenomenon. However, alternative explanations have been proposed in recent research. Studies employing Energy Decomposition Analysis (EDA) have investigated the origins of this directionality. For instance, Stone and colleagues demonstrated that the linearity in Nu···XY halogen bonds is primarily due to exchange repulsion, not electrostatic forces.[8c] Similarly, Scheiner and Adhikari reached the same conclusion for a range of pnictogen, chalcogen, halogen, and hydrogen bonds,[8a] and Hobza and team also observed this phenomenon in halogen bond systems.[8b]

The significance of Pauli repulsion in chemical bonding, in general, has been increasingly recognized over the past decades, beginning with pioneering efforts of Frenking and Bickelhaupt.[10]

Recently, we investigated the origin of directionality in the Bürgi-Dunitz angle using a novel analysis method. This approach, which extends traditional Distortion/Interaction and Energy Decomposition Analysis to a two-dimensional energy surface, has proven highly effective in providing both a qualitative and quantitative understanding of interaction directionality. Its success in elucidating the Bürgi-Dunitz angle has led us to employ this method in our current study, where we aim to unravel the origin of directionality in $H_3N\cdots X_2$ halogen bond systems. The method's demonstrated efficacy in analyzing interaction directionality makes it particularly suitable for this investigation.

Computational Methods

Computational Details

Potential energy surface scans were performed in Gaussian 16 Rev. C.01[11] using ωB97X–D/def2-TZVP.[12] Therefore, relaxed scans were performed using the halogen–nitrogen bond length and the N–X–X angle as variables. The dihedral angle between the X–X bond and one N–H bond was frozen to either 0 or 180° to obtain eclipsed or staggered geometries respectively. Energy decomposi-

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tion analysis was performed in ADF (2023.101)[13] using Gaussian 16 calculated structures. ωB97X–D4/TZ2P was used with a numerical quality of VeryGood and no frozen core.[14] Distortion/Interaction Analysis was performed in ADF as part of the Energy Decomposition Analysis.

2D Distortion/Interaction and Energy Decomposition Analysis

The Distortion/Interaction Analysis, also known as the Activation/Strain Model, is a technique for analyzing chemical interactions.[15] This method was independently developed by the groups of F. Matthias Bickelhaupt and Kendall N. Houk. It dissects the energy of interaction between two entities into two main components. The distortion energy, ΔE_{dist}, quantifies the energy required to modify the isolated relaxed geometry of the fragments to their interaction geometry. The interaction energy, ΔE_{int}, reflects the energy released when these distorted fragments are combined.

The interaction energy can further be dissected into physically meaningful components using Energy Decomposition Analysis.[16] In our research, we utilize the canonical EDA, based on the Ziegler-Rauk Energy Decomposition Analysis.[17] This approach breaks down ΔE_{int} into three parts: the Orbital Interaction Energy (ΔE_{oi}), which accounts for stabilization from intermolecular filled-orbital/empty-orbital interactions and intrafragment polarization; the Pauli Repulsion (ΔE_{Pauli}), describing destabilization due to intermolecular filled-orbital/filled-orbital, which essentially contribute to steric repulsion.; and ΔV_{elstat}, representing the electrostatic attraction between the charge distributions of the interacting fragments.

The Distortion/Interaction Analysis and Energy Decomposition Analysis have been effectively applied in various contexts, from transition state analysis[18] to exploring the nature of covalent[19] and noncovalent[7a,20] interactions or analyzing conformational preferences.[21] Traditionally, these analyses are conducted at defined stationary points, like minima or transition states, to facilitate comparisons across different systems. However, this approach can sometimes yield data that is challenging to interpret or may lead to misleading conclusions.[15,22] This is primarily because these stationary points can occur at varying intermolecular distances in different systems. As the intermolecular distance reduces, all energy components generally increase, complicating direct comparisons. To address this, the analysis is now often performed along reaction coordinates or at "consistent geometries," where energy components at equivalent distances are compared.[15,23] This modification has significantly improved the reliability of these analysis methods.[18h,22a,24]

However, when examining the directionality of interactions, it is advantageous to consider not just a couple of potential interaction angles, but a broad spectrum of them. To address this need, we have recently introduced the 2D Distortion/Interaction and Energy Decomposition Analysis.[25] In this method, the analysis is extended across a two-dimensional potential energy surface (PES), providing a thorough and detailed understanding of the interaction. This approach allows for both qualitative and quantitative insights, offering a more comprehensive view than traditional methods.

Results and Discussion

We chose $H_3N\cdots X_2$ with $X_2 = I_2$, Br_2, Cl_2, or F_2 as halogen bond systems for our study. These systems have long served as an archetypal system for studying halogen bonds.[7a,26] Figure 1 shows the calculated halogen bonded complexes. As expected,

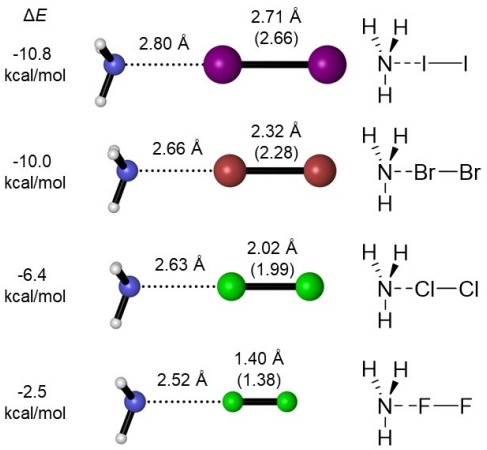


Figure 1. Halogen bond complexes. Distance in parenthesis describes the equilibrium distance of the isolated dihalogen. Energies calculated at the ωB97X–D4/TZ2P//ωB97X–D/def2-TZVP level of theory.

bonding strength increases in the order of F_2, Cl_2, Br_2, and I_2. X–X distances slightly increase upon binding.

For each of the systems a 2D potential energy surface was constructed. The dihalogen was arranged along the x-axis, with the halogen bonding atom in the origin. NH_3 was positioned in a grid defined by the distance to the origin and the angle between the X_2 axis and the $X\cdots NH_3$ interaction. The distance was varied from 2 Å to 4 Å while the angle was varied from 70° to 290°. Here angles below 180° describe an eclipsed confirmation while angles above 180° describe the corresponding staggered confirmation (Figure 2).

The $H_3N\cdots I_2$ System

We initially focused our investigation on the $H_3N\cdots I_2$ system. In Figure 3 the potential energy surface of this complex is illustrated. For an alternative visual representation, please refer

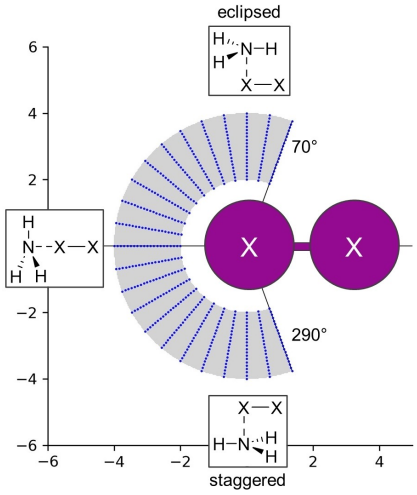


Figure 2. Configuration of 2D surfaces of $H_3N\cdots X_2$ halogen bond systems.

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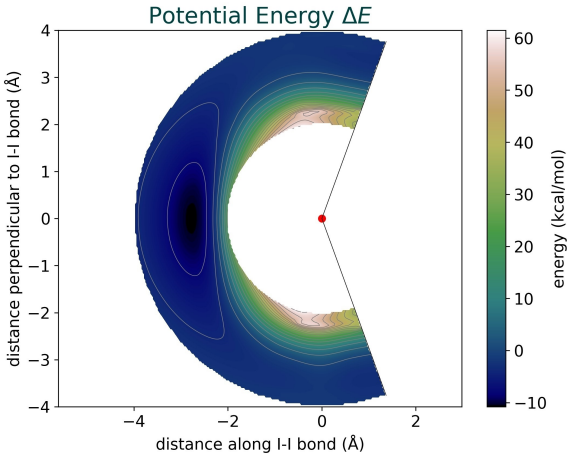


Figure 3. Potential energy surface for the $H_3N\cdots I_2$ system calculated using ωB97X–D4/TZ2P//ωB97X–D/def2-TZVP.

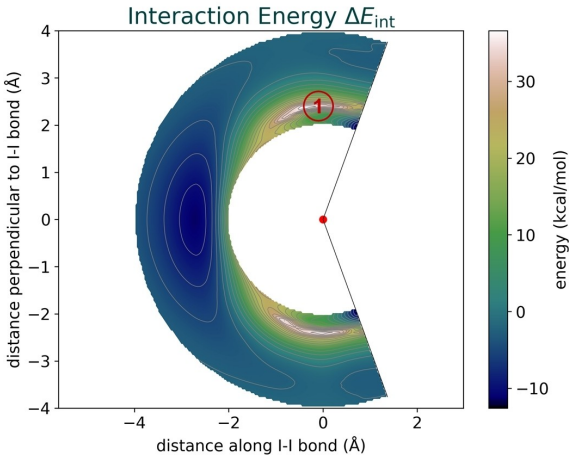


Figure 5. Interaction energy surface for $H_3N\cdots I_2$. Labeled with ① is a high energy region found at low and high angles and at a distance of 2.5 Å.

to the Supporting Information where Figure 3–7, 10, 12, and 14 are presented in a different color scheme.

This surface reveals a distinct energy minimum at a linear configuration, marked by a distance of 2.80 Å. Along the linear axis, the energy exhibits a significant increase when approaching closer distances, reaching a maximum of 21.3 kcal/mol, attributable to steric repulsion between atoms. Notably, the PES demonstrates a pronounced angular dependency, with high-energy regions occurring at angles around 100° and 260°. In these regions, energy levels rise to above 50 kcal/mol. By applying Distortion/Interaction Analysis across this surface, we were able to generate separate surfaces representing both the distortion and the interaction energies.

The distortion energy (Figure 4) remains near zero across most of the surface. However, it shows high energy regions at both low and very high angles, and when approaching closer distances. As expected, the interaction energy (Figure 5) is most favorable at 2.80 Å in the linear configuration, shaping the potential energy surface at the relevant distances, as distortion energy is constant and minimal above 2.4 Å at all angles.

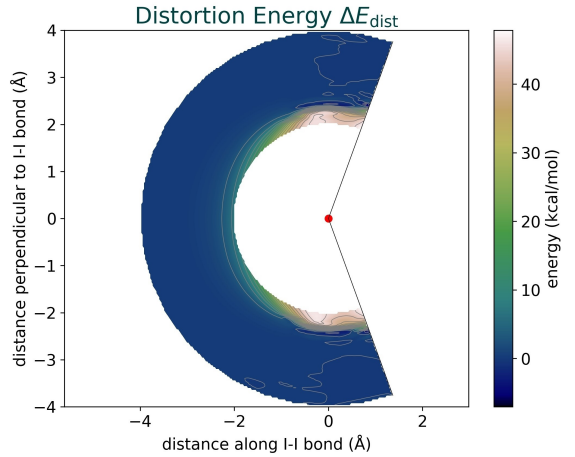


Figure 4. Distortion energy surface for the $H_3N\cdots I_2$ system.

Notably the interaction energy increases strongly when the distance decreases to 2.5 Å at angles around 100° and 260°, but this increase in energy is quickly offset when the nucleophile comes closer than 2.5 Å, as shown in Figure 5, indicated by point ①.

Further examination of the molecular structures in this region reveals the reason for this behavior. Figure 6 represents the I–I distance across the observed surface, now plotted in a polar coordinate system. The I–I distance ranges between 2.65 and 2.80 Å in most areas. It is observed that coordination of NH_3 from a mostly linear (180°) direction leads to slight elongation and weakening of the I_2 bond. However, at angles below 110° and above 230° the I–I distance increases significantly at shorter distances, indicating a transition into an $H_3NI^+\cdots I^-$ complex.

At point 1 (Figure 5), the $H_3N\cdots I_2$ complex remains intact. However, when the NH_3 molecule is pushed even closer to the I_2 dissociation occurs, resulting in the formation of a $H_3NI^+\cdots I^-$ complex. This leads to a decrease in unfavorable interaction energy and an increase in distortion energy. The distortion energy plot essentially mirrors the trends observed in Figure 6 (Supporting information Figure S1).

Using 2D Energy Decomposition Analysis, we further analyze the interaction energy surface (Figure 7). As shown earlier, the interaction energy becomes highly unfavorable

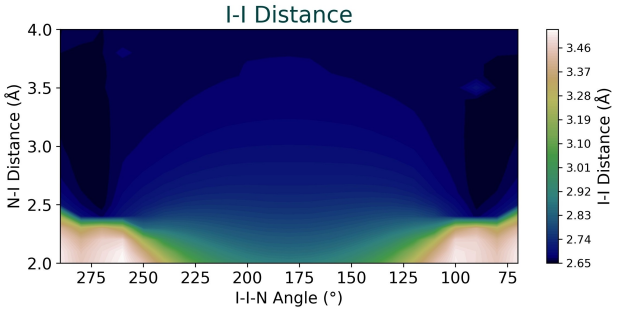


Figure 6. I–I distances across the observed 2D surface.

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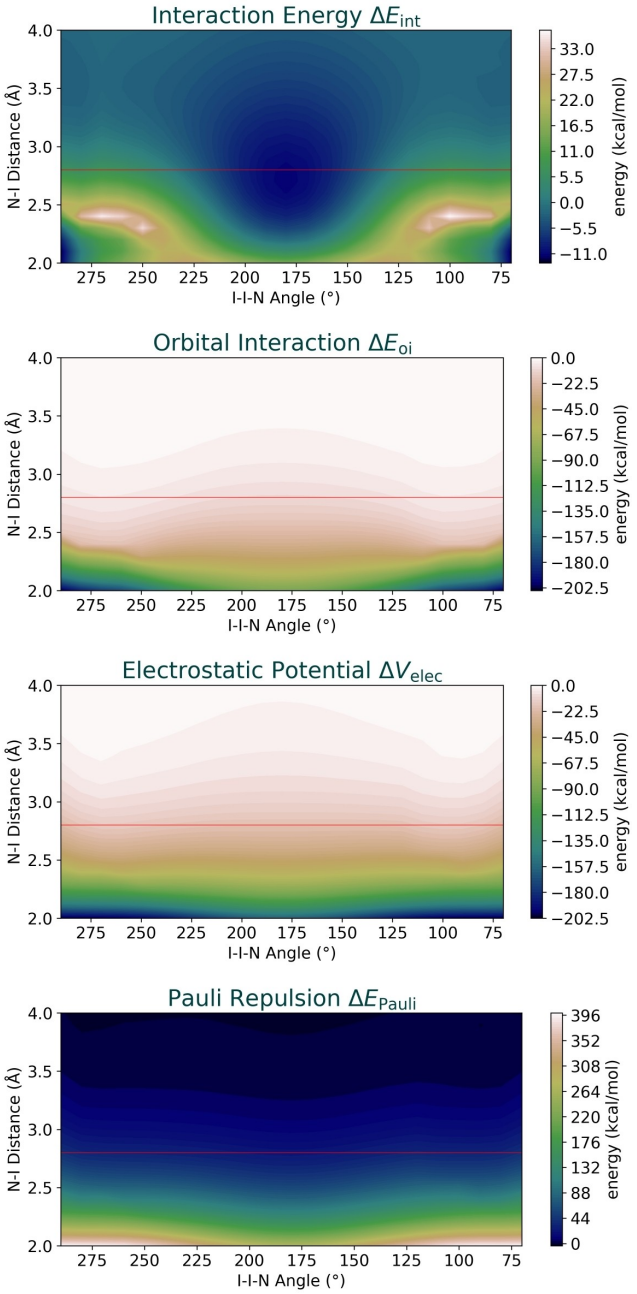


Figure 7. 2D Energy Decomposition Analysis energy surfaces of the $H_3N\cdots I_2$ system. Red lines indicate the equilibrium distance of the complex.

around angles of 100° and 260° at a distance of 2.5 Å due to increased Pauli repulsion. Remarkably, at distances less than 2.4 Å the interaction energy decreases significantly by over 10 kcal/mol. This falloff is caused by a strong and rapid rise in orbital interaction energy. The reason for this strongly increased orbital interaction is that NH_3 now interacts with what has effectively become an I^+ ion after dissociation of the I–I bond, which is a much stronger electrophile. Notably, when a rigid scan is performed with the I–I bond frozen at 2.66 Å, this pattern in orbital interaction is not observed (Supporting Information Figure S2).

At greater distances, e.g., the equilibrium distance for the $H_3N\cdots I_2$ system of 2.80 Å, the orbital interaction is most pronounced at 180° and falls off at higher or lower angles. This can be attributed to the ideal alignment of the I_2 LUMO at 180°, leading to the strongest n–σ* interaction.

Electrostatic interactions display an angular dependence at distances close to 2.8 Å with a slight reduction in these interactions as the angle deviates from 180°. Similarly, Pauli repulsion exhibits its lowest values when the system is in a linear configuration. As the angles decrease or increase from this linear state, Pauli repulsion begins to rise.

In an additional quantitative analysis, we extracted the EDA energy values at a distance of 2.80 Å across angles from 90° to 270°, as shown in Figure 8. As the distortion energy does not play a significant role at this distance, interaction energy directly shapes the potential energy surface and can therefore be analyzed instead. As expected, the staggered configuration (towards an angle of 270°) is favored over the eclipsed configuration. Therefore, the factors leading to the linear configuration should be primarily judged on the behavior of the energy components in the range of angles above 180°. This data indicates that Pauli repulsion plays a notable role in favoring the 180° linear configuration, being less unfavorable at 180° by 9.2 kcal/mol. The orbital interaction energy also contributes to the directionality, favoring the linear arrangement over angled configurations by approximately 5.5 kcal/mol. The electrostatic interaction energy remains relatively constant between 150° and 210°, with an increase of 4.6 kcal/mol towards 270°.

To elucidate why ΔE_{Pauli} demonstrates a strong preference for the linear configuration, we analyzed the electron density around I_2, as illustrated in Figure 9. A significant anisotropy in the electron distribution around the iodine atoms is visible. We observed that the electron density along the I–I axis diminishes more rapidly than in the perpendicular direction. This asymmetry leads to reduced Pauli repulsion when the halogen bond aligns linearly, thereby contributing to the directionality of the bond.

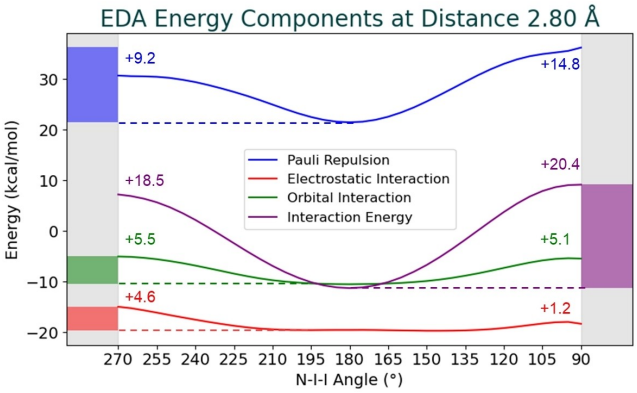


Figure 8. Representation of Energy Decomposition Analysis energy components at a distance of 2.80 Å. The shaded areas in the graph indicate the difference between the lowest and highest values with the value indicating the energetic preference for the linear configuration over the angled.

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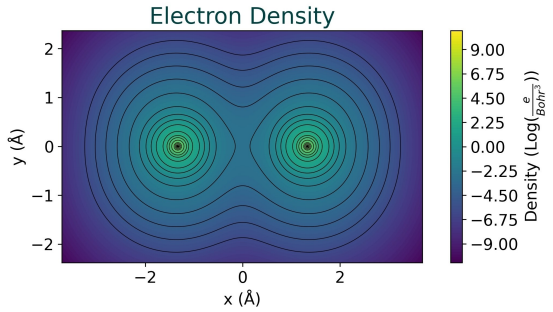


Figure 9. Electron density of the I_2 molecule. Calculated at the ωB97X–D/def2-TZVP level of theory.

From these findings, we conclude that while electrostatic interactions are the main factor to the overall strength of halogen bonding, thus underlining its electrostatic nature, the pronounced directionality of these interactions is not predominantly governed by electrostatic forces, as the σ-hole model suggests. Instead, it is the Pauli repulsion that critically influences the energy surface to favor a linear configuration. Orbital and electrostatic interactions also support this linear arrangement, but their influence is less significant.

The $H_3N\cdots Br_2$ System

The behavior of the $H_3N\cdots Br_2$ complex is strikingly similar to that of the $H_3N\cdots I_2$ complex. Figure 10 displays the potential energy surface of the $H_3N\cdots Br_2$ interaction. Similar to the iodine complex, the high energy regions at extreme angles and close distances are indicative of significant Pauli repulsion, followed by a dissociation of the Br–Br bond into a $NH_3Br^+\cdots Br^-$ complex. The comprehensive 2D Distortion/Interaction and Energy Decomposition Analysis for this complex is shown in the Supporting Information (Figure S3).

Figure 11 focuses on the analysis of energy components at a bonding distance of 2.66 Å in the $H_3N\cdots Br_2$ halogen bond

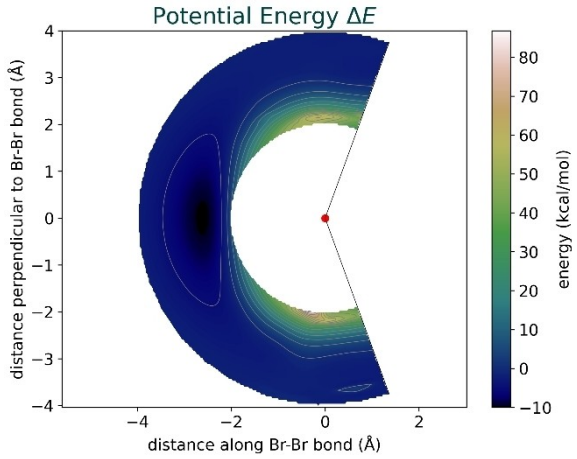


Figure 10. Potential energy surface for the $H_3N\cdots Br_2$ system calculated using ωB97X–D4/TZ2P//ωB97X–D/def2-TZVP.

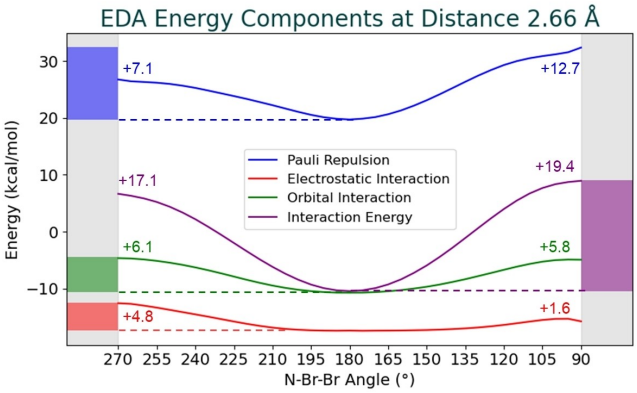


Figure 11. Energy Decomposition Analysis energy components at a distance of 2.66 Å. The shaded areas in the graph represent differences between the lowest and highest values. The given values illustrate the difference between the energy at the staggered configuration at 270° relative to 180°, and the energy at the eclipsed configuration at 90° relative to 180°. This highlights the preference for the linear geometry over both the staggered and eclipsed conformations.

complex. Again, there is a notable difference between the eclipsed (approaching 90°) and staggered (approaching 270°) configurations with the latter being favored. In particular ΔE_{Pauli} and ΔV_{elstat} show significant differences in these two conformations. In the eclipsed conformation, Pauli repulsion is increased due to N–H/I–I interactions. However, this is largely offset by a stronger electrostatic interaction. The orbital interactions appear more symmetrical. The overall shape of the interaction energy is still predominantly defined by Pauli repulsion, although with 7.1 kcal/mol less than for iodine. The preference of Pauli repulsion for a linear configuration can again be explained through analysis of the electron density of the Br_2 molecule, which exhibiting characteristic anisotropy similar to that in the I_2 system (Figure S4). Orbital interactions now also show a strong influence with a 6.1 kcal/mol increase towards 270° while the preference in terms of electrostatics is slightly increased to 4.8 kcal/mol.

The $H_3N\cdots Cl_2$ System

The potential energy surface of the $H_3N\cdots Cl_2$ complex closely resembles that observed in the $H_3N\cdots Br_2$ complex, as shown in Figure 12. Consistent with the patterns seen in the other complexes, the Cl–Cl bond undergoes dissociation when the ammonia is brought close to the dihalogen at either low or high angles. A comprehensive 2D Distortion/Interaction and Energy Decomposition analysis for the $H_3N\cdots Cl_2$ complex is available in the Supporting Information (Figure S5).

The Energy Decomposition Analysis of the energy components at a bonding distance of 2.63 Å in the $H_3N\cdots Cl_2$ complex, shown in Figure 13, reveals characteristics different to those observed before. In the staggered configuration ΔE_{Pauli} is not the strongest contributor to the directionality anymore, but rather orbital interactions, closely followed by electrostatics. This can be explained by the smaller size of chlorine, exerting

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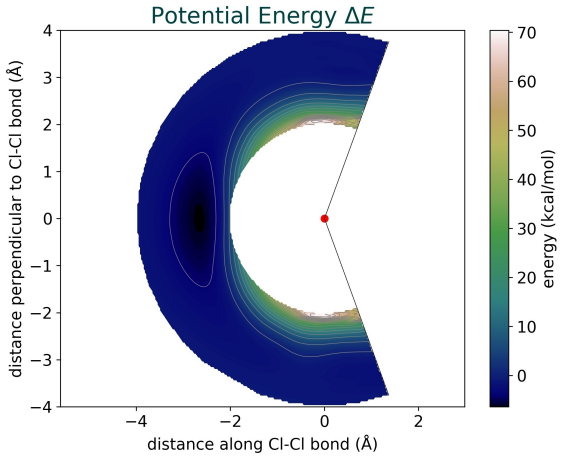


Figure 12. Potential energy surface for the $H_3N\cdots Cl_2$ system calculated using ωB97X–D4/TZ2P//ωB97X–D/def2-TZVP.

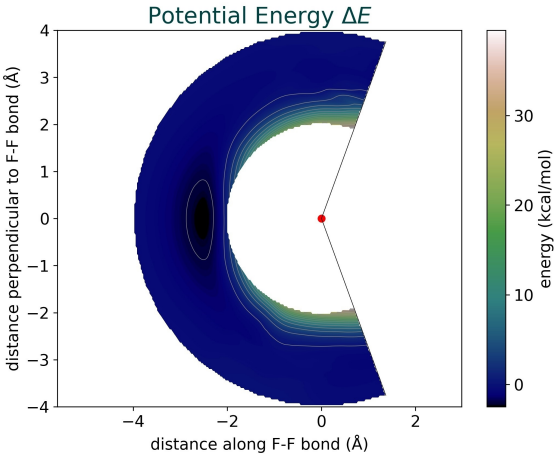


Figure 14. Potential energy surface for the $H_3N\cdots F_2$ system calculated using ωB97X–D4/TZ2P//ωB97X–D/def2-TZVP.

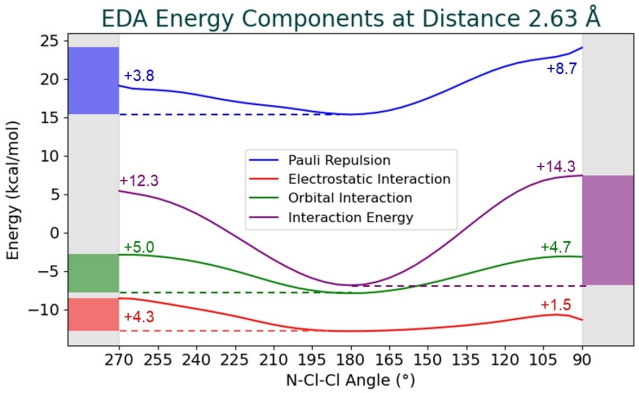


Figure 13. Energy Decomposition Analysis energy components at a distance of 2.63 Å. The shaded areas in the graph represent differences between the lowest and highest values. The given values illustrate the difference between the energy at the staggered configuration at 270° relative to 180°, and the energy at the eclipsed configuration at 90° relative to 180°. This highlights the preference for the linear geometry over both the staggered and eclipsed conformations.

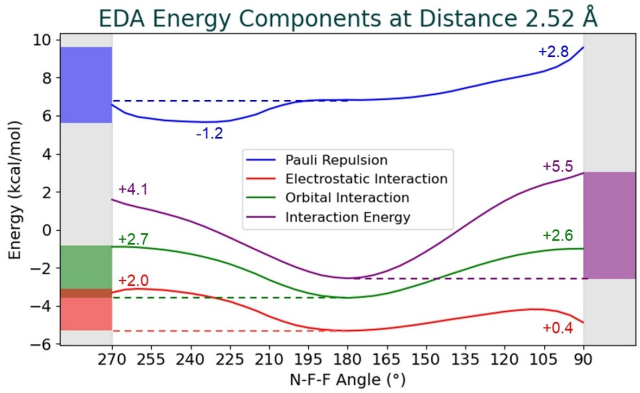


Figure 15. Energy Decomposition Analysis energy components at a distance of 2.52 Å. The shaded areas on the graph indicate the differences between the lowest and highest values. The provided values illustrate the differences in energy at 180° compared to the staggered configuration at 270°, and then compared to the eclipsed configuration at 90°. It is important to note that for Pauli repulsion, the reference value in the eclipsed configuration was not the value at 270°, but rather the lowest value found in the region greater than 180°, due to Pauli repulsion favoring an angle of approximately 235°.

lower Pauli repulsion compared to bromine, at a similar distance.

The $H_3N\cdots F_2$ System

The $H_3N\cdots F_2$ complex forms a notably weaker bond compared to the other halogen complexes. Despite this, its potential energy surface, as shown in Figure 14, qualitatively resembles that of the other halogens. A key difference, however, is the absence of F–F bond dissociation across the observed surface in the $H_3N\cdots F_2$ complex. This is likely due to the smaller size and stronger bond of F_2 compared to the other halogens.

When analyzing the Energy Decomposition Analysis energy components at a bonding distance of 2.52 Å for the $H_3N\cdots F_2$ complex, as depicted in Figure 15, the results differ significantly from those of the other complexes.

In the $H_3N\cdots F_2$ complex, unlike the others, the 180° angle is not uniformly favored across all energy components. Here, Pauli repulsion shows a minimum in the staggered configuration. However, the electrostatic interaction, which counterbalances this trend, is least favorable in this same region. Interestingly, electrostatic interactions are most favorable at both 90° and 180° angles. Nevertheless, in the eclipsed configuration, Pauli repulsion increases substantially. The orbital interactions in the $H_3N\cdots F_2$ complex are fairly symmetrical and now play the predominant role in shaping the interaction energy in the staggered region, closely followed in influence by electrostatics, while Pauli repulsion would favor an angled conformer.

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Conclusions

In conclusion, our study utilized multi-dimensional Distortion/Interaction and Energy Decomposition Analysis to investigate the halogen bonds of I_2, Br_2, Cl_2, and F_2 with NH_3. For the complexes involving I_2, Br_2, and Cl_2, we observed that all EDA energy components demonstrate a preference for the linear configuration. In the I_2 complex, the most significant factor influencing this preference is a decrease in Pauli repulsion in the linear arrangement, which can be attributed to the anisotropy in the electron density of dihalogens. As we examine smaller halogens, the impact of ΔE_{Pauli} is progressively reduced. In the Cl_2 complex, orbital interactions emerge as the dominant contributor to the directionality with electrostatic interactions also playing a significant role. In the case of the F_2 complex, an interesting shift occurs. While Pauli repulsion in F_2 might suggest a preference for a non-linear geometry, it is the combined forces of both orbital interactions and electrostatics that ultimately drive the system towards a bond angle of 180°.

Our findings align with those of other research groups, suggesting that the directionality of halogen $H_3N\cdots X_2$ systems is not necessarily driven by electrostatic interactions caused by the σ-hole. Instead, it appears to be a result of a combination of factors, including orbital interactions and electrostatic forces, with the most substantial influence being the reduced Pauli repulsion in the linear configuration in the bigger halogens. However, we also demonstrate that the size of the halogen plays a crucial role, as ΔE_{Pauli} loses influence as we move to smaller halogen bond acceptors. This comprehensive understanding of the interaction provides a more nuanced perspective on the nature of halogen bonding in these systems.

Supporting Information

Additional Computational Analyses.

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: Halogen Bond · Energy Decomposition Analysis · Directionality · Distortion/Interaction Analysis · Pauli Repulsion

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